BLOCKCHAIN-BASED AGRI-PRODUCT TRACEABILITY WITH DATA VALIDATION AND SUSTAINABILITY ASSESSMENT

A PROJECT REPORT

Submitted by

MOHAMED ARSATH A	**2116231901030**
UDAYA SANKAR C	**2116231901058**

in partial fulfillment for the award of the degree

of

BACHELOR OF ENGINEERING

in

COMPUTER SCIENCE AND ENGINEERING (CYBER SECURITY)





RAJALAKSHMI ENGINEERING COLLEGE

ANNA UNIVERSITY: CHENNAI 600 025

OCTOBER 2026

RAJALAKSHMI ENGINEERING COLLEGE

CHENNAI

BONAFIDE CERTIFICATE

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SIGNATURE

Dr. P. Tamilselvi, Ph.D.,

HEAD OF THE DEPARTMENT

PROFESSOR

Department of Computer Science and Engineering (Cyber Security)

Rajalakshmi Engineering College

Chennai – 602 105

SIGNATURE

Dr. V. Balasubramanian, Ph.D.,

SUPERVISOR

PROFESSOR

Department of Computer Science and Engineering (Cyber Security)

Rajalakshmi Engineering College

Chennai – 602 105

Submitted to Project Viva-Voce Examination held on _____

Internal Examiner **External Examiner**

ABSTRACT

Agricultural products pass through several stages before reaching consumers, including cultivation, harvesting, processing, transportation, storage, and retail. Information generated at these stages is often maintained using separate systems, making it difficult to establish a reliable end-to-end product history and increasing the possibility of incorrect or manipulated records. Blockchain-based traceability improves record integrity, but does not guarantee the correctness of data before it is recorded.

This project proposes a blockchain-based agricultural product traceability system that introduces a data validation stage before records are committed to a permissioned blockchain network. The system maintains batch-level information throughout the supply chain and uses role-based access control for authorized stakeholders. Before an event is recorded, it is checked for completeness, authorization, duplication, and predefined business rules.

A QR code is generated for each agricultural batch, allowing consumers to retrieve product information by scanning the code. The system also captures selected sustainability indicators and combines their normalized values into a batch-level sustainability score.

The proposed framework integrates data validation, blockchain-based traceability, QR-based consumer verification, and sustainability assessment. The prototype will be evaluated using validation, blockchain performance, QR verification, and sustainability metrics.

ACKNOWLEDGEMENT

First, we thank the almighty God for the successful completion of the project. Our sincere thanks to our Chairman **Mr. S. Meganathan B.E., F.I.E.,** for his sincere endeavor in educating us in his premier institution. We would like to express our deep gratitude to our beloved Chairperson **Dr. Thangam Meganathan Ph.D.,** for her enthusiastic motivation which inspired us a lot in completing this project and Vice Chairman **Mr. Abhay Shankar Meganathan B.E., M.S.,** for providing us with the requisite infrastructure.

We also express our sincere gratitude to our college Principal **Dr. S.N. Murugesan M.E., Ph.D.,** and **Dr. P. Tamilselvi Ph.D., Professor and Head,** Department of Computer Science and Engineering (Cyber Security) and our project guide **Dr. V. Balasubramanian, Professor,** for his encouragement and guiding us throughout the project. We would like to thank our project coordinator **Mr. Benedict J.N., M.E., Assistant Professor (SG),** for his encouragement and guidance towards successful completion of this project and to our parents, friends, all faculty members and supporting staffs for their direct and indirect involvement in the successful completion of this project, and for their encouragement and support.

MOHAMED ARSATH A 2116231901030

UDAYA SANKAR C 2116231901058

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CHAPTER 1

INTRODUCTION

1.1 BACKGROUND

Agricultural products pass through several stages before reaching the final consumer, including farming, harvesting, processing, transportation, storage, and retail. At each stage, information such as crop type, batch number, harvesting date, quantity, farm location, processing details, transportation conditions, and storage parameters is generated. In conventional supply chains, this information is often maintained through paper records, spreadsheets, or independent databases. Since the records are distributed across different participants, establishing a complete and reliable history of an agricultural batch can be difficult.

Blockchain technology provides a potential solution by maintaining a distributed ledger in which accepted records are cryptographically linked and difficult to modify retrospectively. This makes blockchain useful for maintaining the integrity and traceability of supply-chain records. However, blockchain does not guarantee that the information entered into the ledger is correct. An incorrect quantity, temperature reading, or product detail can become a permanent part of the ledger if it is accepted without proper verification.

To address this limitation, the proposed project introduces a data validation layer between data collection and blockchain storage. Before an agricultural event such as harvest registration, processing, transportation, storage, or retail is recorded, the system checks required fields, user authorization, duplicate batch information, valid ranges, and predefined business rules. Only records that satisfy the applicable validation conditions are forwarded to the permissioned blockchain for storage.

The system also uses QR codes to connect physical agricultural products with their digital traceability records. Consumers can scan the QR code to access available information about the product and its supply-chain history. In addition, sustainability-related parameters such as water usage, fertilizer usage, fuel consumption, transportation distance, and energy consumption can be associated with each batch and combined into a normalized sustainability score.

The proposed system therefore combines data validation, blockchain-based traceability, QR-based consumer verification, and batch-level sustainability assessment within a single framework. The emphasis on role-specific interaction and simple QR-based access is intended to make the system practical for different supply-chain participants while improving the transparency and reliability of agricultural product information.

Taken together, the proposed system brings four major functions into a single, coherent framework:

1. Agricultural batch registration and lifecycle management.
2. Pre-recording data validation.
3. Blockchain-based, tamper-evident traceability.
4. QR-based consumer verification combined with sustainability assessment

1.2 PROBLEM STATEMENT

Agricultural supply chains involve multiple independent participants, each of whom may record information using different tools, formats, and conventions. This fragmentation creates several specific problems that this project aims to address:

- Lack of a common, shared traceability mechanism spanning every stage of the supply chain, from farm to retail shelf.

- Possibility of incomplete or internally inconsistent product information being accepted into whatever record-keeping system a participant happens to use.
- Difficulty in identifying data submissions made by unauthorized or unverified participants.
- Possibility of duplicate or conflicting batch records entering the system, whether by accident or intentionally.
- Limited and inconsistent means for an ordinary consumer to verify the claims made about a product's origin and handling.
- A weak or entirely absent connection between traceability records and the sustainability footprint of the product they describe.
- Heavy dependence on centralized databases operated by individual organizations, which introduces a single point of failure and a single point of potential manipulation.
- Difficulty in reconstructing a clear, chronological history of a product's movement once it has changed hands several times.

The project therefore focuses on developing a system in which incoming supply-chain data is validated before it is ever recorded on a permissioned blockchain, while QR codes provide a simple and accessible mechanism for both internal stakeholders and end consumers to retrieve and verify that information.

1.3 OBJECTIVES

The overall objective of this project is to design and develop a blockchain-based agricultural product traceability system that measurably improves the reliability and transparency of supply-chain information compared to fragmented, unvalidated record-keeping approaches currently in common use. The specific objectives pursued in this work are:

1. To register agricultural products using unique, system-generated batch identifiers.
2. To collect structured information from each distinct stage of the agricultural supply chain.
3. To validate all supply-chain data for completeness, format, and business-rule compliance before it is written to the blockchain.
4. To detect and reject incomplete, inconsistent, duplicate, or unauthorized records at the point of submission.
5. To maintain validated batch information using a permissioned blockchain network.
6. To use smart contracts to enforce which supply-chain operations each role is permitted to perform.
7. To generate and manage QR codes associated with individual agricultural product batches.
8. To allow consumers to verify available product and supply-chain information through a simple scan-based interface.
9. To calculate a batch-level sustainability score from selected, measurable environmental indicators.
10. To provide a transparent, auditable, end-to-end history of each agricultural product from registration to retail.

1.4 SCOPE

The scope of this project covers the traceability of agricultural products from initial batch registration to consumer verification. The system considers the following participants:

- Farmer
- Processor

- Warehouse Operator
- Distributor
- Retailer
- Administrator
- Consumer

The system maintains the following information throughout the batch lifecycle:

- Batch identifier and product/crop name
- Farmer details and production location
- Production and harvest dates
- Quantity at different stages
- Processing and transportation details
- Storage information
- Sustainability-related parameters
- Verification status

The current phase focuses on batch-level traceability using a permissioned blockchain network. The system includes data validation, blockchain-based record management, QR-based consumer verification, and sustainability assessment.

The following are outside the scope of the current phase:

- Individual-item-level tracking
- Public, permissionless blockchain deployment
- Fully automated sensor-based data collection
- Regulatory environmental certification

- Payment or financial settlement between participants

The sustainability component provides a relative, indicator-based assessment using selected measurable parameters and is not intended to serve as formal environmental certification.

Beyond these functional requirements, the system is also governed by a set of non-functional requirements that shape its design, summarized in the table below.

Requirement	Description	Design Implication
Performance	Validation and blockchain write operations should complete within an interactive time budget appropriate for a web application	Permissioned, not public, blockchain
Scalability	The system should support a growing number of batches and stakeholders without redesign	Batch-level rather than item-level tracking
Availability	Consumer-facing verification should remain accessible even under moderate load	Off-chain read cache for common queries
Usability	Non-technical stakeholders (farmers, consumers) should not need blockchain expertise	QR-based interface, plain-language validation errors
Maintainability	Validation rules and sustainability weights should be updatable without redeploying the ledger	Rules and weights configured outside chaincode logic

Table 1.1 Non-Functional Requirements

1.5 SECURITY REQUIREMENTS / SECURITY CONSIDERATIONS

Because agricultural supply-chain information has downstream consequences for food safety, commercial trust, and consumer decision-making, it should not be freely modifiable by unauthorized participants. The following security requirements guide the design of the proposed system.

1. Authentication

All system users must authenticate before performing any operation. Each participant uses an individual account based on their assigned role.

2. Role-Based Authorization

Access to system functions is controlled according to the participant's role.

Role	Main Permission
Farmer	Register crop/batch information
Processor	Add processing information
Distributor	Add transportation information
Warehouse	Add storage information
Retailer	Update retail-stage information
Admin	Manage users and monitor records
Consumer	View verified product information (read-only)

Table 1.2 Role-Based Permission Matrix

3. Data Integrity

Validated records stored on the permissioned blockchain are maintained as tamper-evident records, providing an auditable history of batch activities.

4. Smart Contract Control

Smart contracts enforce the permissions associated with each participant and restrict unauthorized supply-chain operations.

5. Duplicate Prevention

The system checks batch identifiers and rejects duplicate registrations to maintain a unique record for each batch.

6. Input Validation

Important fields such as batch ID, quantity, dates, location, and other submitted information are checked for valid format and consistency before being recorded.

7. Auditability

Each recorded event contains relevant information such as the participant, timestamp, batch ID, operation, and supply-chain stage to support traceability and auditing.

8. QR Security

QR codes reference the corresponding batch identifier and allow users to retrieve the associated traceability information without directly exposing sensitive data.

9. Data Privacy

Sensitive business information is restricted to authorized participants, while only appropriate product and verification information is made available through the consumer-facing QR interface.

1.6 SDG ALIGNMENT

This project can be associated with the following United Nations Sustainable Development Goals, reflecting both its technical contribution and its intended real-world impact.

S.No.	SDG No. & Goal	Relevant SDG Target	Project Contribution / Alignment
1	SDG 2 – Zero Hunger	Target 2.4: Sustainable food production systems	Supports reliable agricultural supply-chain information and batch-level product traceability, reducing losses attributable to mismanagement and undetected spoilage.
2	SDG 9 – Industry, Innovation and Infrastructure	Target 9.c: Access to information and communication technology	Applies blockchain, smart contracts, and QR-based digital infrastructure to modernize agricultural supply-chain management.
3	SDG 12 – Responsible Consumption and Production	Target 12.8: Access to relevant information for sustainable development	Provides consumers with verified product information and quantified sustainability indicators to support informed purchasing decisions.
4	SDG 13 – Climate Action	Target 13.3: Improve awareness on climate mitigation	Incorporates transport distance, fuel consumption, and energy usage into a visible, batch-level sustainability score.

Table 1.3 SDG Alignment

CHAPTER 2

LITERATURE REVIEW

2.1 EXISTING SYSTEM WITH SECURITY APPROACHES

Existing agricultural traceability systems can be broadly classified into centralized database, QR-based, IoT-based, and blockchain-based approaches. Each approach provides different levels of traceability, security, and data reliability.

2.1.1 Centralized Traceability Systems

Centralized systems store supply-chain information in a database managed by a central organization. They are relatively simple to implement but depend on the availability and security of the central authority. Unauthorized modification or compromise of the central database can affect the reliability of stored records.

2.1.2 QR-Based Traceability

QR codes provide a simple way to connect a physical agricultural product with its digital information. Consumers can scan the code to access product or batch details. However, QR codes mainly provide an access mechanism and do not ensure the accuracy or integrity of the information stored behind them.

2.1.3 Blockchain-Based Traceability

Blockchain-based systems maintain supply-chain records in a distributed ledger shared among participating organizations. Transactions form a chronological and tamper-evident history of product activities. Permissioned platforms such as Hyperledger Fabric are suitable for controlled supply-chain environments because participant identities and access permissions can be managed.

2.1.4 Summary of Security Approaches in Existing Systems

Existing approaches commonly use authentication, role-based access control, cryptographic techniques, smart contracts, permissioned blockchain networks,

sensor monitoring, and QR-based identification. However, the level of integration between data validation, blockchain traceability, consumer verification, and sustainability assessment varies across existing systems. This provides the basis for identifying the research gap addressed by the proposed system.

2.1.5 Review of Recent Blockchain-Based Agricultural Traceability Literature (2024–2026)

Recent studies show that blockchain is increasingly used for improving traceability and transparency in agricultural supply chains. Alanazi [1] proposed a permissioned blockchain framework with smart contracts and QR-based consumer verification across major supply-chain stages. Other studies [2], [3], and [6] demonstrate the use of blockchain, IoT, and smart contracts for improving food safety, traceability, and supply-chain management.

Studies [4], [5], and [7] highlight important challenges such as data quality, standardization, access control, and regulatory requirements. Reviews [8]–[13] further identify issues including stakeholder adoption, infrastructure limitations, interoperability, cost, scalability, and limited real-world deployment.

Recent works [14], [15] address batch-level traceability and blockchain scalability, while studies [16] and [17] extend blockchain applications toward supply-chain risk management and QR-based anti-counterfeiting. These studies demonstrate that blockchain, smart contracts, and QR verification provide a practical foundation for agricultural traceability.

However, the reviewed literature shows limited integration of pre-recording data validation with blockchain traceability and quantified batch-level sustainability assessment. Therefore, the proposed system combines data validation, permissioned blockchain, QR-based consumer verification, and sustainability assessment in a single framework.

Ref.	Focus Area	Relevance to Proposed System
[1]	Permissioned blockchain, QR verification	Base architecture; lacks pre-ledger validation and sustainability scoring
[2],[3]	Blockchain + IoT for food safety	Confirms feasibility of permissioned blockchain in agri-food
[4]	Evaluation framework for traceability	Motivation for validation layer; evaluation metrics
[5]	Blockchain characteristics for agri-food	Justifies immutability + automation as design priorities
[6],[7]	Smart contracts; redactable data schemes	Informs role-based access and smart-contract design
[8]–[10]	Global trends, feasibility, adoption	Practical deployment and adoption considerations
[11]	Digital transformation, data quality	Direct motivation for pre-recording validation
[12],[13]	Sustainability, scoping review	Motivates grounded sustainability-scoring methodology
[14],[15]	Batch-level traceability; storage optimization	Informs batch data model and scalability design
[16]	Risk management via blockchain	Reinforces tamper-evidence as core requirement
[17]	Dual-QR anti-counterfeiting	Informs live-query QR verification design
[18]	Agriculture supply chain blockchain	Confirms maturity of base pattern; validates research gap

Table 2.1 Summary of Reviewed Literature (2024–2026)

2.1.6 Comparative Discussion

The reviewed studies [4], [5], [11]–[13] show that blockchain improves the integrity and traceability of agricultural records after they are stored. However, ensuring the correctness of data before it is recorded remains a challenge. This highlights the need for a data validation layer before blockchain storage, which is a key focus of the proposed system.

Another common pattern is the use of QR codes to provide simple consumer access to complex blockchain-based traceability systems [1], [17]. The proposed system follows the same approach by allowing consumers to verify batch information through a QR code without interacting directly with the blockchain infrastructure.

The literature also shows that sustainability is increasingly associated with agricultural traceability, but few systems provide a reproducible batch-level sustainability score [8], [12], [13]. Therefore, the proposed system extends

blockchain traceability by integrating sustainability indicators with individual agricultural batch records.

Overall, the comparison identifies two major gaps: pre-recording data validation and integrated batch-level sustainability assessment. These form the main focus of the proposed system.

2.2 RESEARCH GAPS

From the literature reviewed above, five specific gaps motivate the design of the proposed system.

Gap 1 — Validation Before Blockchain Recording

Blockchain protects data after recording, but incorrect input data may still become part of the ledger. The proposed system therefore validates data before blockchain storage.

Gap 2 — Traceability and Sustainability Treated Separately

Existing systems often treat sustainability separately from product traceability. The proposed system links sustainability information directly to individual agricultural batches.

Gap 3 — Limited Consumer-Level Verification

Some systems focus mainly on organizational traceability. The proposed system provides consumers with simple QR-based verification of batch information.

Gap 4 — Fragmented Rather Than Integrated Architecture

Data collection, validation, blockchain storage, QR verification, and sustainability assessment are often addressed separately. The proposed system integrates these functions into a single workflow.

Gap 5 — Permission Structure for Known Supply-Chain Participants

Agricultural supply chains involve identifiable participants with different responsibilities. The proposed system therefore uses a permissioned blockchain with role-based access rather than an open public blockchain.

2.3 PROPOSED SYSTEM

2.3.1 Overview of the Proposed System

The proposed system starts with the registration of an agricultural batch by an authorized participant. The submitted data is first checked by the data validation module for completeness, correctness, duplication, and predefined rules. Invalid records are returned for correction, while valid records are stored on a permissioned blockchain using a unique batch ID.

The same validation process is applied to subsequent events such as processing, transportation, storage, and retail. A QR code linked to the batch ID is generated for consumer verification. Consumers can scan the QR code to view the available traceability information and the batch's sustainability score, calculated from selected environmental indicators collected throughout the supply chain.

2.3.2 Objectives of the Proposed System

The proposed system aims to:

- Improve traceability of agricultural products across the supply chain.
- Validate data before it is recorded on the blockchain.
- Provide role-based access for supply-chain participants.
- Maintain a complete and auditable batch history.
- Enable consumers to verify product information using QR codes.
- Associate sustainability information with individual batches.
- Improve transparency among supply-chain participants.
- Develop a practical prototype using available technologies.

2.3.3 Advantages of the Proposed System

1. Improved traceability — product information can be followed continuously across multiple supply-chain stages rather than reconstructed after the fact.
2. Pre-recording validation — data is checked for correctness and consistency before it ever becomes part of the immutable ledger.
3. Tamper-evident history — the blockchain layer provides an auditable record of every accepted transaction, with any retroactive alteration immediately detectable.
4. Controlled participation — permissioned access prevents unknown or unverified users from directly interacting with the blockchain network.
5. Consumer verification — QR codes provide a simple, low-friction way for consumers to access batch information without any blockchain-specific knowledge.
6. Sustainability assessment — measurable environmental indicators are associated with individual batches rather than reported only in aggregate.
7. Auditability — every supply-chain activity is tied to a timestamp, an actor identity, and a transaction record, supporting investigation and accountability when needed.

CHAPTER 3

SYSTEM DESIGN

3.1 SYSTEM ARCHITECTURE

The proposed system consists of six layers: User, Application, Data Validation, Permissioned Blockchain, Output, and Consumer Verification.

The User Layer includes farmers, processors, distributors, warehouses, retailers, administrators, and consumers. The Application Layer provides login, batch registration, event entry, QR generation, and management functions.

The Data Validation Layer checks submitted data for completeness, correctness, duplicate records, user authorization, and predefined rules before blockchain storage. Valid data is stored in the Permissioned Blockchain Layer, where smart contracts enforce role-based operations.

The Output Layer generates QR codes and calculates the batch sustainability score. The Consumer Verification Layer allows consumers to scan the QR code and view verified batch information.

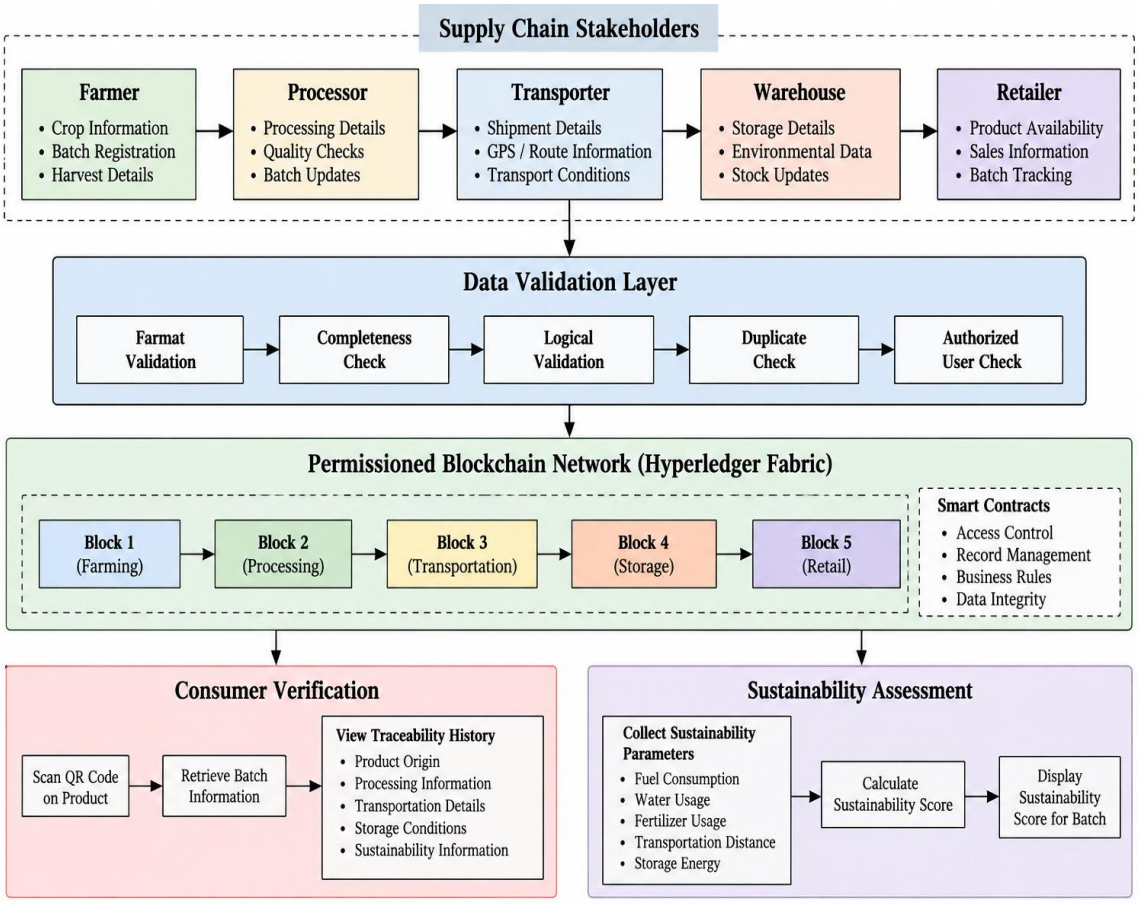


Fig. 3.1 System Architecture

3.2 MODULE DESIGN

The system is divided into functional modules for batch management, data validation, blockchain traceability, QR verification, and sustainability assessment.

Module 1 — User and Role Management

Manages user accounts and assigns role-based permissions for supply-chain activities.

Module 2 — Agricultural Batch Management

Creates and manages batches using details such as batch ID, product, farmer, location, quantity, and harvest date.

Module 3 — Data Validation

Checks completeness, format, authorization, duplicates, and business rules before allowing data to enter the blockchain.

Module 4 — Blockchain Traceability

Stores validated batch events such as registration, processing, transportation, storage, and retail in the permissioned blockchain.

Module 5 — Smart Contract Management

Uses smart contracts to enforce role-based permissions for different supply-chain participants.

Module 6 — QR Code Generation and Verification

Generates a QR code linked to the batch ID, allowing consumers to access verified product information.

Module 7 — Sustainability Assessment

Calculates a batch-level sustainability score using normalized environmental indicators such as water, fertilizer, fuel, transport, and energy usage.

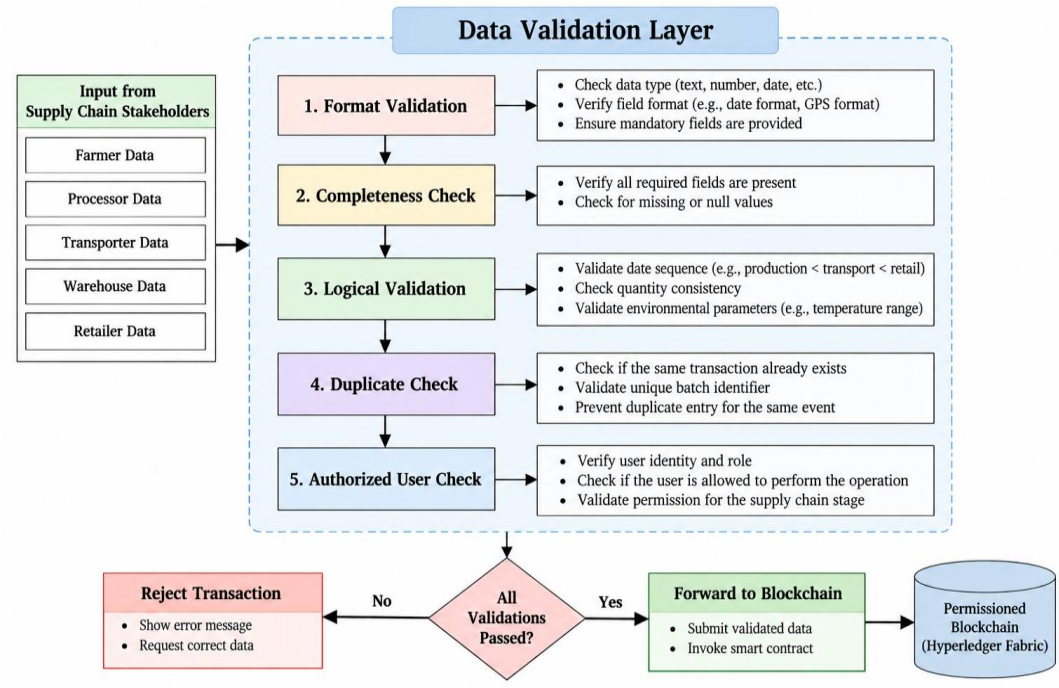


Fig. 3.2 Pre-Recording Data Validation Process

3.3 DATA FLOW / UML DIAGRAMS

This section presents the use case, activity, and sequence diagrams to describe the system's users, workflow, and component interactions. The data flow consists of batch registration, data validation, blockchain recording, QR generation, QR verification, and sustainability assessment, supported by the blockchain ledger and off-chain storage for larger documents.

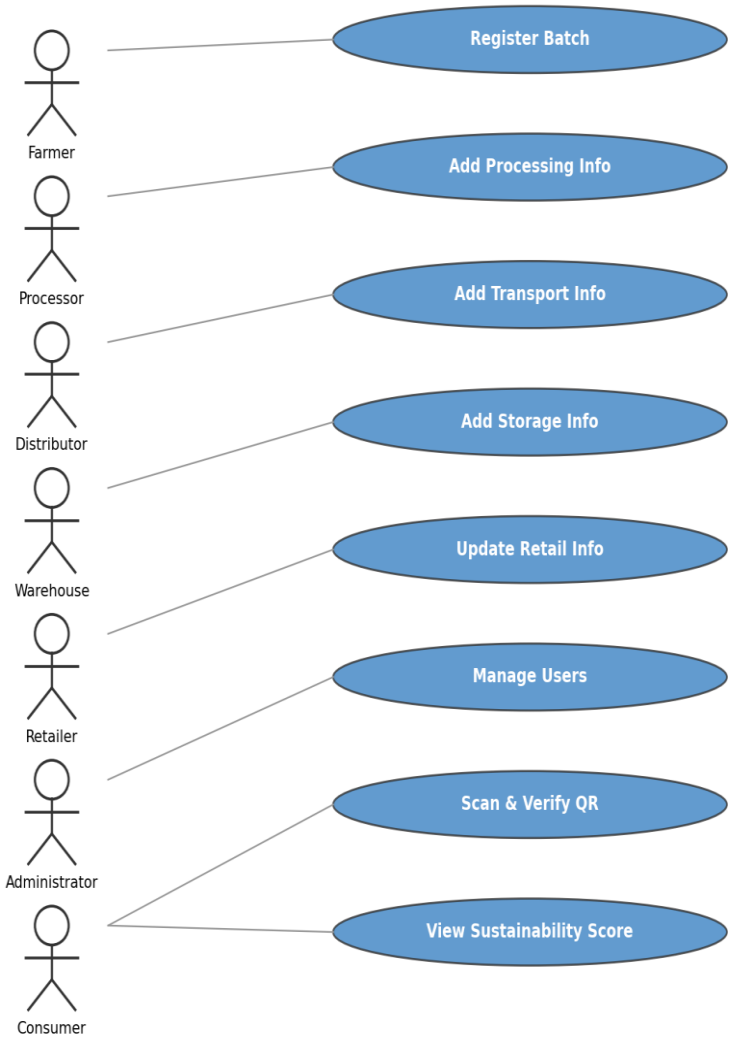


Fig. 3.3 Use Case Diagram

Fig. 3.3 shows the seven system actors and their main use cases. Write operations are restricted through role-based access control, while consumers can scan QR codes and view verified batch and sustainability information.

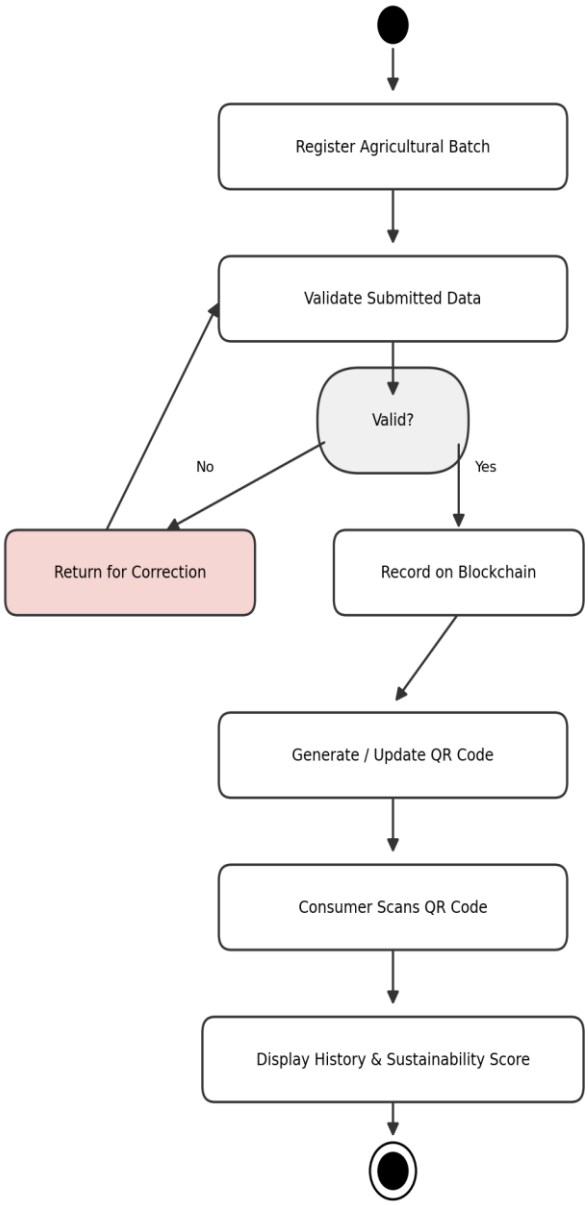


Fig. 3.4 Activity Diagram

Fig. 3.4 shows the flow of a supply-chain event from submission through validation and blockchain recording. Invalid data is returned for correction, while valid data proceeds to blockchain storage, QR update, and consumer display with the batch sustainability score.

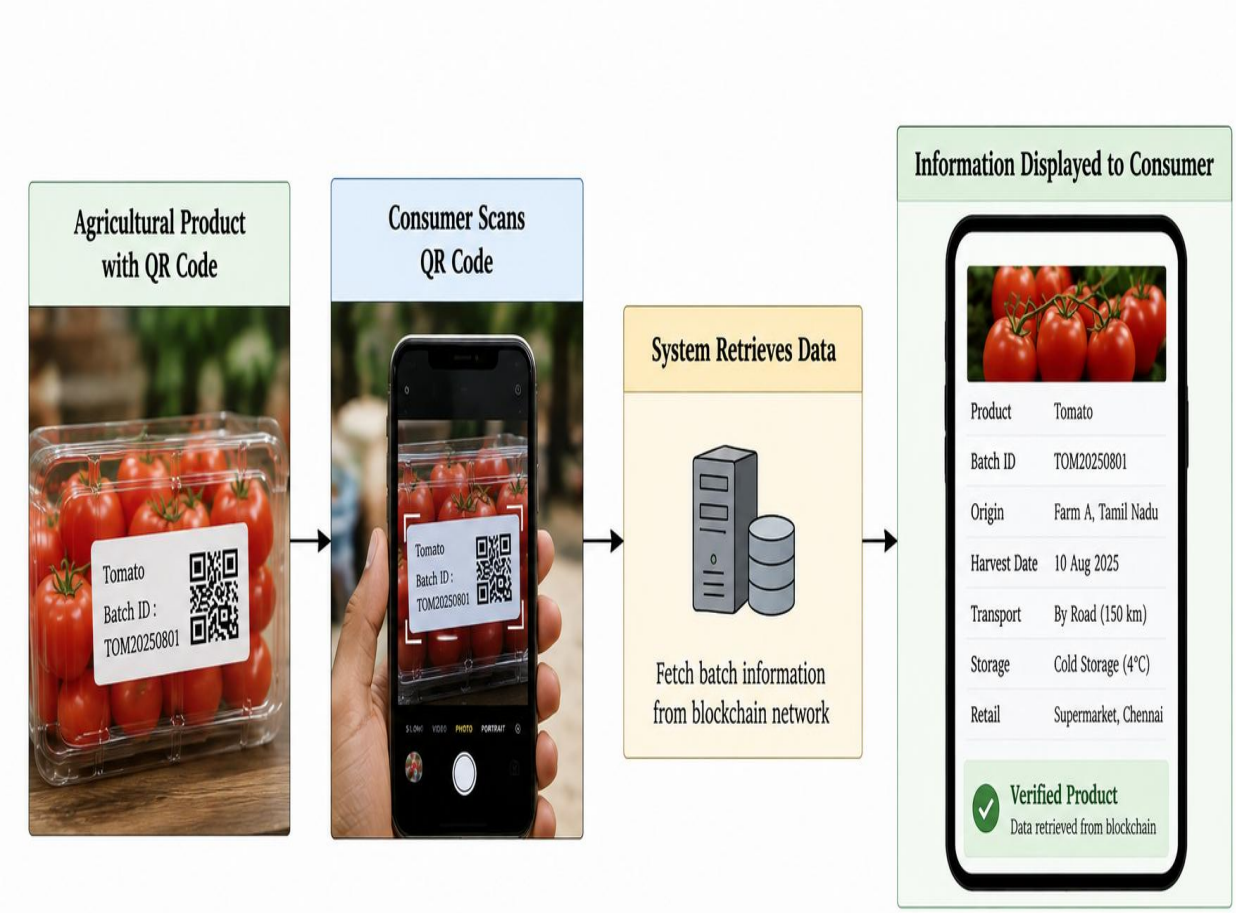


Fig. 3. QR-based consumer verification of an agricultural product batch.

Fig. 3.5 QR-Based Consumer Verification

Fig. 3.5 shows the consumer verification process, where scanning the QR code retrieves the corresponding batch information from the permissioned blockchain and displays the verified product record to the consumer.

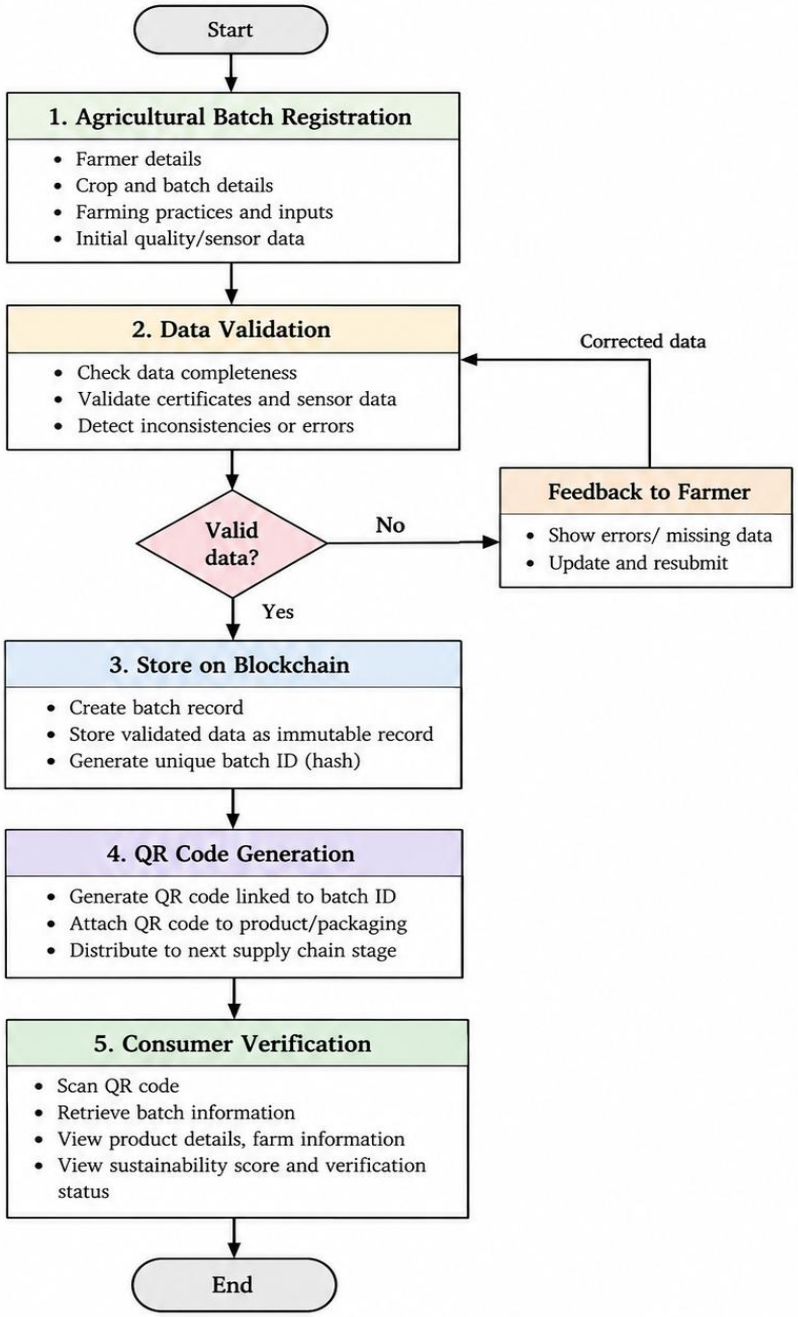


Fig. 5. End-to-end workflow from agricultural batch registration to consumer verification.

Fig. 3.6 End-to-End Project Workflow

Fig. 3.6 shows the complete system workflow from batch registration and data validation to blockchain storage, QR generation, and consumer verification, including the correction loop for invalid data.

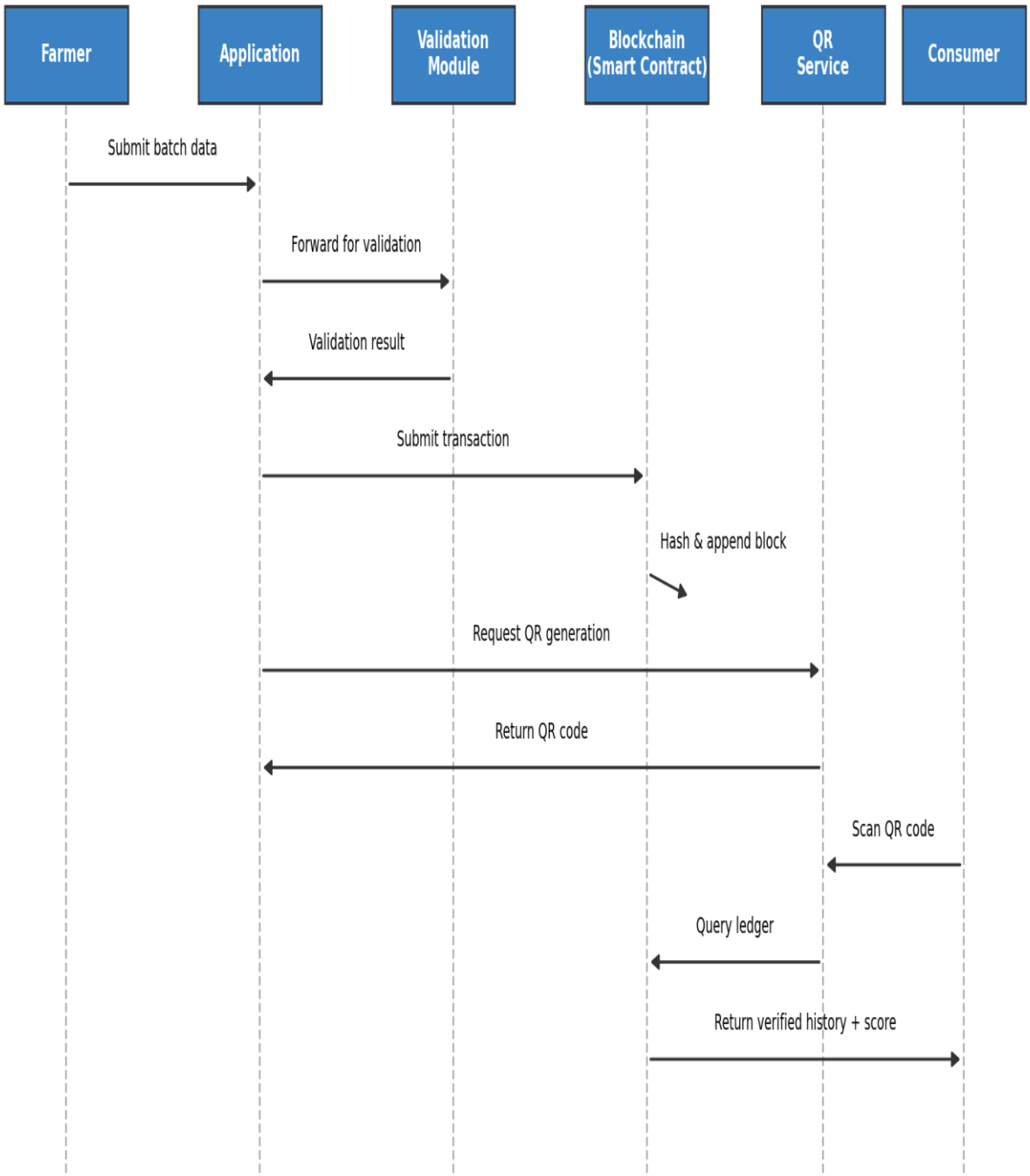


Fig. 3.7 Sequence Diagram

Fig. 3.7 shows the message flow between the main system components, where a QR scan triggers a fresh blockchain query to retrieve the latest verified batch information.

3.4 THREAT MODEL AND ATTACK SURFACE

The main threats identified in the proposed system are:

1. **Unauthorized Login** – Attackers may attempt to access accounts using stolen or guessed credentials.
2. **Unauthorized Transactions** – Users may attempt operations beyond their assigned roles.
3. **False Data Submission** – Authorized users may submit incorrect batch information.
4. **Duplicate Batch Registration** – The same batch may be registered more than once.
5. **QR Manipulation** – Fraudulent QR codes may be placed on legitimate products.
6. **API Attacks** – Malicious requests may target the application interface.
7. **Unauthorized Blockchain Access** – Unapproved participants may attempt to access or submit transactions.
8. **Sensor/Data Manipulation** – Manipulated sensor readings may affect traceability and sustainability results.

Threat	Primary Attack Surface	Potential Impact
Unauthorized login	Authentication endpoint	Account takeover
Unauthorized operation	Application API	Invalid or unauthorized ledger entry
False data submission	Data entry forms	Misleading traceability record
Duplicate batch registration	Batch registration endpoint	Ambiguous or conflicting batch history
QR manipulation	Physical product packaging	Consumer misled about product identity
API abuse	Application interface	Service disruption or data exposure
Unauthorized blockchain access	Network membership service	Unverified transaction submission

Table 3.1 Threat Model Summary

3.5 SECURITY CONTROLS

The following controls are proposed to mitigate the threats identified in Section 3.4.

Threat	Security Control
Unauthorized login	Authentication with verified credentials
Unauthorized operation	Role-based access control, enforced in smart contracts
False data	Pre-recording data validation module
Duplicate batch	Unique, system-enforced batch identifiers
Transaction manipulation	Permissioned blockchain with hash-chaining
Unauthorized blockchain access	Network membership / identity-based access control
QR manipulation	Batch-ID verification against live ledger query
API abuse	Input validation and API-level security controls
Data modification	Cryptographic hashing and immutable ledger history
Audit issues	Structured transaction logs with timestamp and actor identity

Table 3.2 Security Controls Mapping

3.6 DESIGN VALIDATION CHECKLIST

As a closing check on the design presented in this chapter, Table 3.5 maps each research gap identified in Section 2.2 directly to the specific module or design element that addresses it, so that the connection between motivation and design is explicit rather than implied.

Research Gap (Section 2.2)	Addressed By
Gap 1 — Validation before blockchain recording	Module 3 (Data Validation)
Gap 2 — Traceability and sustainability treated separately	Module 7 (Sustainability Assessment)
Gap 3 — Limited consumer-level verification	Module 6 (QR Code Generation and Verification)
Gap 4 — Fragmented rather than integrated architecture	Overall six-layer architecture
Gap 5 — Permission structure for known participants	Permissioned blockchain layer

Table 3.3 Research Gap to Design Element Mapping

CHAPTER 4

SYSTEM DESCRIPTION

4.1 DATA COLLECTION

The system collects data from different supply-chain participants, including:

Farmer Data:

- Farmer ID, farm location, crop, cultivation date, harvest date, quantity, and farming inputs.

Processing Data:

- Processor ID, processing date, processing type, quantity after processing.

Transportation Data:

- Source, destination, transportation distance, vehicle information, fuel consumption.

Storage Data:

- Warehouse ID, storage date, temperature, humidity, energy consumption.

Retail Data:

- Retailer ID, retail location, date received, available quantity.

Sustainability Data:

- Water consumption, fertilizer usage, fuel consumption, transportation distance, energy consumption.

4.2 DATA PREPROCESSING

Before any submitted record is stored, it passes through a sequence of preprocessing operations designed to catch errors as early as possible in the pipeline.

Step 1 — Data Cleaning

Identifies missing or invalid values.

Step 2 — Format Validation

Checks dates, quantities, and identifiers for correct formats.

Step 3 — Duplicate Detection

Checks whether the batch or record already exists.

Step 4 — Role Verification

Confirms that the participant is authorized for the requested operation.

Step 5 — Business Rule Validation

Checks conditions such as valid dates, positive quantities, unique batch IDs, authorized event types, and required fields.

Step 6 — Blockchain Preparation

Once a record has passed every preceding check, it is converted into a transaction format suitable for submission to the permissioned blockchain network.

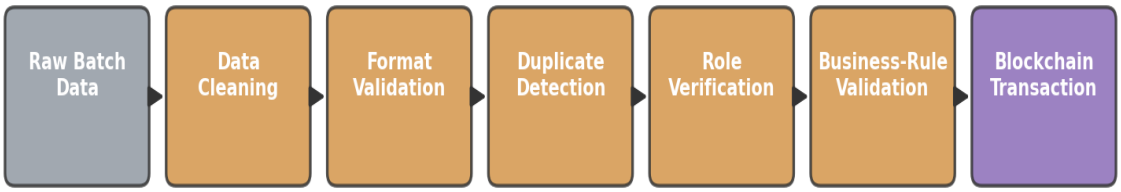


Fig. 4.1 Data Validation and Preprocessing Pipeline

4.2.1 ALGORITHMS / METHODOLOGY

A. Data Validation Algorithm

The system receives the batch record, checks required fields and formats, verifies the user role and batch ID, and applies business rules. Invalid records are returned for correction, while valid records are approved and sent to the blockchain.

B. Blockchain Recording Method

Validated records are stored as structured blockchain transactions:

$$Ri = (Bi, Si, Ei, Ti, Li, Qi, Hi)$$

where Bi is the batch identifier, Si is the supply-chain stage, Ei is the event, Ti is the event timestamp, Li is the location, Qi is the quantity, and Hi is the associated hash or reference information linking this record to the previous one in the chain.

A transaction is recorded only when all validation conditions are satisfied:

$$Tvalid = Vdata \land Vrole \land Vbatch \land Vrule$$

C. Sustainability Score Calculation

Environmental indicators are normalized to a common scale before calculating the sustainability score. For lower-is-better indicators:

$$si = (xi^{max} - xi) / (xi^{max} - xi^{mIn})$$

For higher-is-better indicators:

$$si = (xi - xi^{mIn}) / (xi^{max} - xi^{mIn})$$

The overall score is:

$$S = \Sigma\ (w_i \cdot s_i),\ \text{for } i = 1 \text{ to } n,\ \text{subject to } \Sigma\ w_i = 1$$

The weights and normalization ranges will be finalized during the experimental stage based on the selected agricultural product and available data.

4.2.2 RESULTS AND DISCUSSION

As this is a Phase I report, the section focuses on the current implementation and planned evaluation rather than assumed performance results. The proposed workflow covers batch registration, data validation, blockchain recording, QR verification, and sustainability assessment.

The system ensures that data is validated before being recorded on the blockchain, helping prevent incorrect or inconsistent records. Functional testing will cover batch registration, invalid-data rejection, duplicate detection, role-based access, blockchain transactions, and QR-based retrieval.

The sustainability module will be tested using sample environmental data. Performance evaluation will consider validation time, blockchain transaction latency, ledger query time, and QR verification time.

Module	**Evaluation Metrics**
Data Validation	Validation accuracy, rejection rate, duplicate detection rate, response time
Blockchain	Transaction latency, throughput, query time, transaction success rate
QR Verification	Generation time, scan time, retrieval time, verification success rate
Sustainability	Normalization accuracy, score consistency, batch-level assessment correctness
System (End-to-End)	End-to-end response time and overall functional reliability

Table 4.1 Planned Evaluation Metrics

No final numerical performance values are reported at this stage, as the complete experimental evaluation has not yet been conducted. These values will be measured and reported in the subsequent phase of this project once implementation and testing have been carried out.

4.3 ADDITIONAL ALGORITHMS

D. QR Verification Algorithm

Input: Scanned QR reference (batch identifier or lookup URL)

The QR code is scanned to obtain the batch ID, which is then used to retrieve the corresponding blockchain records. The system verifies the blockchain history by checking the stored hash values. If all hashes match, the batch is marked verified. otherwise, it is marked tamper-detection for administrator review. The current sustainability score is then retrieved and the verified batch information is displayed to the consumer.

4.4 DATABASE AND LEDGER SCHEMA DESIGN

The system maintains two complementary data stores: a permissioned blockchain ledger for validated, tamper-evident events, and an off-chain relational store for user accounts, session data, and large supporting artifacts that are impractical to store directly on-chain.

The relationship between the off-chain store and the blockchain ledger is deliberately asymmetric: the off-chain store can be rebuilt at any time by replaying the blockchain's event history, whereas the blockchain ledger cannot be reconstructed from the off-chain store. This ensures that the blockchain, not the conventional database, remains the single source of truth for traceability, while the database exists purely to make common queries fast and to hold data that does not need, or is too large, to live on-chain.

4.5 TESTING STRATEGY

The proposed system follows a layered testing approach, where each module is tested individually before performing complete end-to-end system testing.

Test Category	Scope	Example Test Cases
Unit Testing	Individual validation rules, scoring formulas, hash functions	Reject negative quantity; reject out-of-range temperature
Integration Testing	Validation → Blockchain → QR pipeline	Valid batch reaches ledger and produces a resolvable QR code
Security Testing	Role enforcement, duplicate prevention, tamper detection	Processor attempting a retailer-only operation is rejected
Regression Testing	Re-running prior test cases after code changes	Previously passing validation rules still pass after updates
Usability Testing	Consumer-facing QR scan and result display	Scan-to-result time and clarity of displayed information

Table 4.2 Testing Strategy Summary

Test data will include both valid records and deliberately invalid records, such as missing fields, out-of-range values, duplicate batch IDs, and unauthorized role attempts.

4.6 DEPLOYMENT CONSIDERATIONS

The proposed system is designed for deployment using Hyperledger Fabric, with participating organizations assigned different roles. Each organization operates its own peer nodes, while an ordering service manages transaction ordering. A shared channel provides all authorized participants with a common batch history.

Endorsement policies ensure that transactions are accepted only from authorized organizations according to their assigned roles. For the current phase, the system is developed and tested using a local Fabric test network with simulated organizations. Future deployment can extend this setup to separate peer and ordering infrastructure for real-world organizations without changing the core application and smart-contract logic.

4.7 LIMITATIONS OF THE CURRENT PHASE

The current phase has several limitations. The validation system uses rule-based and threshold-driven checks and does not yet include advanced anomaly detection. The sustainability scoring formula is defined, but its weights require calibration using data from a specific agricultural product and region.

The Hyperledger Fabric setup is currently a single-host test network with simulated organizations rather than a distributed deployment. Therefore, performance values from existing literature are treated only as reference points. These limitations are considered in the future work planned for Chapter 5.

4.8 REGULATORY AND COMPLIANCE CONSIDERATIONS

Although this project is primarily a technical prototype rather than a compliance deliverable, its data model has been designed with awareness of the kind of recordkeeping that food-safety regulators typically expect. In the Indian context, the Food Safety and Standards Authority of India (FSSAI) requires traceability information connecting a food product back to its source, including production date, batch identity, and handling conditions — all of which correspond directly to fields already present in the batch schema. The validation layer's enforcement of mandatory fields and plausible value ranges is consistent with, though not a substitute for, the kind of data completeness a regulatory audit would expect to find.

The system's audit trail — every event tied to a timestamp, an actor identity, and a cryptographic hash is also the kind of record a food-safety investigation would need during a recall: given a reported problem with a specific batch, an investigator can trace every stage that batch passed through, and verify, via the hash chain, that the record was not altered after the fact. This project does not claim regulatory certification of any kind, but its design does not conflict with the recordkeeping principles that such certification would eventually require.

CHAPTER 5

CONCLUSION

The proposed system combines data validation, permissioned blockchain, QR-based verification, and batch-level sustainability assessment to improve the traceability and transparency of agricultural products. Data is validated before being recorded on the blockchain, helping to reduce incomplete, duplicate, inconsistent, and unauthorized records while maintaining a tamper-evident history of each batch.

The system also enables consumers to access verified product information through QR codes. Sustainability indicators such as water usage, fertilizer usage, transportation distance, fuel consumption, and energy consumption are associated with each batch and used to calculate a sustainability score.

Overall, the proposed approach addresses the identified research gaps while providing a practical and measurable framework for agricultural traceability. Future work will focus on advanced anomaly detection, real-world agricultural data, IoT integration, and distributed multi-organization deployment.

5.1 SUMMARY OF CONTRIBUTIONS

The main contributions of this project are: (i) a pre-recording data validation layer to prevent invalid or unauthorized records; (ii) a permissioned blockchain for secure batch traceability; (iii) QR-based consumer verification using live ledger data; and (iv) a weighted normalization-based method for batch-level sustainability scoring.

5.2 FUTURE WORK

Future work will focus on completing the system modules and integrating them into a single application. The system will then be tested individually and through the complete end-to-end workflow.

The sustainability scoring weights will be calibrated using representative data, and the system will be evaluated using the performance and security metrics defined in the report. Future extensions may include anomaly detection, IoT sensor integration, and distributed multi-organization Hyperledger Fabric deployment.

The current validation system mainly detects incomplete, malformed, duplicate, and out-of-range data. It does not guarantee detection of deliberately false but realistic information. Addressing such cases will require additional data sources and advanced detection techniques. Overall, the project focuses on a practical integration of data validation and sustainability assessment with established blockchain-based agricultural traceability.

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APPENDIX

A. SAMPLE BATCH DATA

Field	Example Value
Batch ID	AGRI2026B001
Product	Rice
Farmer	F001
Location	Thanjavur
Quantity	500 kg
Harvest Date	10-08-2026
Processing	Milling
Transport Distance	85 km
Water Usage	To be recorded during testing
Energy Usage	To be recorded during testing

B. SAMPLE VALIDATION OUTPUT

Valid Batch:

✓ Required fields present ✓ Valid user role ✓ Unique Batch ID ✓ Valid quantity ✓ Valid date ✓ No duplicate detected Status: ACCEPTED

Invalid Batch:

✓ Required fields present ✗ Duplicate Batch ID Status: REJECTED Reason: Batch already exists

C. PROJECT SCREENSHOTS

Selected implementation screenshots and key source-code snippets will be included here once the corresponding modules have been implemented.

D. SAMPLE SMART CONTRACT PSEUDOCODE

The following pseudocode illustrates the structure of the registerBatch and addTransportInfo smart-contract functions referenced in Table 3.2, showing how role verification and validation are enforced at the chaincode level before a transaction is committed to the ledger.

```
function registerBatch(batchId, productName, farmerId, location, quantity,
harvestDate):     if caller.role != "FARMER":         return
error("UNAUTHORIZED")     if ledger.exists(batchId):         return
error("DUPLICATE_BATCH_ID")     if quantity <= 0 or harvestDate >
currentDate():         return error("INVALID_FIELD")     record = { batchId,
productName, farmerId, location, quantity, harvestDate,
previousHash: null, currentHash: hash(record) }     ledger.put(batchId, record)
return success(batchId)  function addTransportInfo(batchId, distance, fuelUsed,
destination):     if caller.role != "DISTRIBUTOR":         return
error("UNAUTHORIZED")     batch = ledger.get(batchId)     if batch == null:
return error("BATCH_NOT_FOUND")     if distance <= 0 or fuelUsed < 0:
return error("INVALID_FIELD")     event = { batchId, type: "TRANSPORT",
distance, fuelUsed, destination,              previousHash: batch.currentHash,
currentHash: hash(event) }     ledger.appendEvent(batchId, event)     return
success(event.currentHash)
```

D. PROJECT RISK REGISTER

The following risks have been identified for the remaining phases of this project, along with the planned mitigation approach for each.

Risk	Likelihood	Mitigation
Hyperledger Fabric network configuration delays implementation timeline	Medium	Use documented Fabric test-network samples as a starting configuration
Sustainability weighting scheme lacks sufficient calibration data	Medium	Use published reference ranges from reviewed literature as interim weights
Validation rules too strict, rejecting legitimate edge-case data	Low	Review rejected-record logs during testing and adjust thresholds
Team unfamiliarity with chaincode development in Hyperledger Fabric	Medium	Allocate initial development time to guided chaincode tutorials
QR verification performance degrades under concurrent scans	Low	Introduce an off-chain read cache for frequently scanned batches

Table 4.4 Project Risk Register

E. SAMPLE CONSUMER-FACING VERIFICATION OUTPUT

The following illustrates the information a consumer is expected to see after scanning a batch's QR code, once the complete workflow described in Chapter 3 has been implemented.

Field	Displayed Value
Product	Tomato
Batch ID	AGRI2026B014
Farm Location	Coimbatore
Harvest Date	02-09-2026
Processing	Washing & Grading — Verified
Transportation	45 km — Verified
Storage	6 °C, 3 days — Verified
Retail	Received 06-09-2026 — Verified
Blockchain Status	VERIFIED — no tampering detected
Sustainability Score	0.78 / 1.00 (example, pending final weighting)

the sustainability score shown above is illustrative of the intended display format only, it is not a measured result from the current implementation.

F. SAMPLE VALIDATION ERROR CODES

The following error codes are returned by the data validation module described when a submitted record fails a specific check, allowing the front-end application to display a precise, actionable message to the submitting stakeholder rather than a generic failure notice.

Error Code	Triggering Condition	Example Message Shown to User
ERR_MISSING_FIELD	A mandatory field is empty or absent	“Harvest date is required.”
ERR_INVALID_FORMAT	A field does not match its expected format	“Quantity must be a positive number.”
ERR_DUPLICATE_BATCH	The submitted batch identifier already exists	“Batch ID already registered.”
ERR_UNAUTHORIZED_ROLE	The user's role does not permit this operation	“Only distributors can add transport data.”
ERR_OUT_OF_RANGE	A numeric or sensor value falls outside the plausible range	“Temperature value is outside the expected range.”
ERR_SEQUENCE_VIOLATION	An event is submitted out of the expected stage order	“Cannot add retail data before storage data.”

G. SAMPLE SUSTAINABILITY INDICATOR REFERENCE RANGES

The following illustrative reference ranges, to be refined during the calibration step described ,give a starting point for normalizing the sustainability indicators defined in Section 4.2.1.C for a typical vegetable crop.

Indicator	Unit	Typical Low (Good)	Typical High (Poor)
Water Usage	L/kg	150	500
Fertilizer Usage	kg/kg	0.05	0.30
Transportation Distance	km	20	300
Storage Energy	kWh/kg	0.10	0.80
CO2 Emissions (estimated)	kg CO2/kg	0.20	1.50

A batch whose indicators fall near the “Typical Low” column across the board would be expected to score toward the high end of the sustainability band described , while a batch near the “Typical High” column would score toward the

low end; the exact boundary values will be fixed once representative data for a specific crop and region has been collected, as noted in Section 4.7.

It is worth noting that these reference ranges are drawn from general agricultural sustainability literature rather than from measurements specific to the crops this project's test batches will use; one of the concrete, measurable outputs expected from the calibration step in Section 4.7 is a revised version of this table populated with values observed from actual test data rather than external reference figures.

H. SAMPLE ROLE-BASED LOGIN FLOW DESCRIPTION

The following describes the expected login and landing-page behaviour for each role, to be implemented as part of Module 1 (User and Role Management).

Role	Post-Login Landing View	Primary Action Available
Farmer	My Batches dashboard	Register new batch
Processor	Pending processing queue	Add processing record
Distributor	Active shipments list	Add transport record
Warehouse	Storage inventory view	Add storage record
Retailer	Received batches list	Update retail record
Administrator	System monitoring dashboard	Manage users, view logs
Consumer	QR scan screen (no login required)	Scan and view batch history

All roles except consumers authenticate through the application gateway and receive a role-based session token. Each API request carries this token, while the smart contract independently verifies the user's role before allowing write operations.

The consumer role requires no authentication, allowing users to access product traceability information directly through QR-based verification.

I. LIST OF ABBREVIATIONS

Abbreviation	Expansion
API	Application Programming Interface
DFD	Data Flow Diagram
QR	Quick Response (Code)
RBAC	Role-Based Access Control
SDG	Sustainable Development Goal
SHA-256	Secure Hash Algorithm, 256-bit
TPS	Transactions Per Second
UML	Unified Modeling Language
UTAUT	Unified Theory of Acceptance and Use of Technology